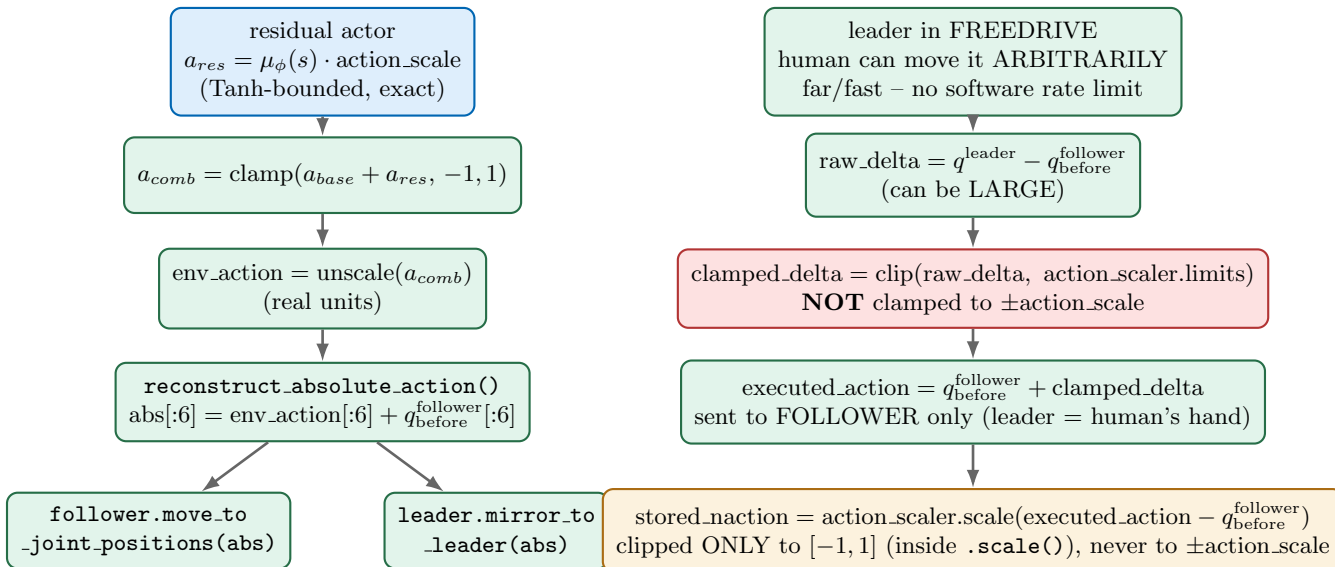


PANEL A – autonomous tick: leader stays
sync'd by construction

PANEL B – intervened: human drives leader,
follower chases (clamped)



*SAME absolute target abs, computed ONCE from the follower's OWN freshly-measured state, sent to both arms every tick – no independent delta accumulation on either side, so any tracking lag self-corrects the next tick rather than compounding.
stored_naction = a_comb (the full combined action).*

stored_naction is BYTE-IDENTICAL to what's physically executed (same clamp applied before both) – critic never associates a clipped/wrong action with a next-state reached via a bigger, unclamped move.

Does a large intervention-derived action “corrupt” training? No – verified against

REAL_RESIDUAL_RL_HUMAN_INTERVENTION_PLAN.md (pre-existing, explicit design decision, not an oversight found here):

1. The buffer's action field is ALWAYS the full combined/executed action, never an isolated residual. The critic is trained as $Q_\theta(s, a)$ via ordinary TD-learning on whatever valid $a \in [-1, 1]^7$ actually occurred – exactly the same as any other off-policy transition. It has no notion of “this action came from a human” or “this exceeds what the actor could propose” – both are irrelevant to fitting Q .

2. The residual actor's OWN output is independently, architecturally bounded to $\mu_\phi(s) \cdot \text{action_scale}$ where $\mu_\phi(s) \in [-1, 1]^7$ is Tanh-squashed (actor.py: build_fc(...) ends in nn.Tanh()) – this bound holds regardless of what values happen to exist in the replay buffer. A large stored intervention action cannot “push” the actor's own output range; there is no mechanism that tries to.

3. No BC/imitation loss exists for the residual actor. BC_LOSS_COEF=0.0 in this .sh, and the design doc notes .compute_actor_bc_loss() asserts not self.residual_actor – that code path is dead here. The actor is trained ONLY by Q-maximization over its own (bounded) proposals – it is never regressed toward matching a stored action value directly, so it cannot be asked to “reproduce” something outside action_scale.

4. Net effect: a large human-correction action simply teaches the critic $Q(s, a_{\text{large}})$ for a region of action space the actor doesn't itself visit – extra, valid supervision, not corruption. This is the same tolerance already documented for offline BC-derived residual targets elsewhere in this codebase (debug_offline/pct_target_exceeds_scale – a metric on a DIFFERENT, offline-BC-actor training path that doesn't run for residual actors, but reflecting the identical underlying principle: exceeding action_scale is expected/tolerated, the actor just saturates through its Tanh).